

FLASK-K8S-DEMO

app.py

Dockerfile

requirements.txt

deployment.yaml

service.yaml

OUTLINE

TIMELINE

Dockerfile

requirements.txt

deployment.yaml

service.yaml

```
< app.py >
1 from flask import Flask, jsonify
2 from datetime import datetime
3
4 app = Flask(__name__)
5
6 @app.route("/")
7 def home():
8     return jsonify({
9         "message": "Hello from Flask API!",
10        "status": "success",
11        "timestamp": datetime.now().isoformat()
12    })
13
14 @app.route("/api/data")
15 def get_data():
16     data = {
17         "id": 1,
18         "name": "Docker Kubernetes Demo",
19         "technology": "Flask + Docker + Kubernetes"
20     }
21     return jsonify(data)
22
23 @app.route("/api/health")
24 def health():
25     return jsonify({"status": "healthy"})
26
27 if __name__ == '__main__':
28     app.run(host='0.0.0.0', port=5000)
```

flask==2.3.3

deployment.yaml

service.yaml

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4   name: flask-api
5 spec:
6   replicas: 2
7   selector:
8     matchLabels:
9       app: flask-api
10  template:
11    metadata:
12      labels:
13        app: flask-api
14    spec:
15      containers:
16      - name: flask
17        image: sahanaa44/flask-k8s-demo:1.0
```

```
1 apiVersion: v1
2 kind: Service
3 metadata:
4   name: flask-service
5 spec:
6   type: NodePort
7   selector:
8     app: flask-api
9   ports:
10   - port: 80
11     targetPort: 5000
12     nodePort: 30080
```

TERMINAL

```
PS C:\Users\sahanaa\flask-k8s-demo> docker build -t sahanaa44/flask-k8s-demo:
:1.0 .
[+] Building 10.2s (0/0) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> [internal] load .dockerignore                                                  0.0s
=> [internal] load context: 28                                                    0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim              1.6s
=> [1/3] FROM docker.io/library/python:3.11-slim@sha256:cd...                  1.6s
=> [2/3] WORKDIR /app                                                            0.0s
=> [3/3] COPY . .                                                                0.1s
=> exporting to image                                                            0.2s
=> exporting layers                                                              0.2s
=> writing image sha256:7e3c8b9f.....                                           0.0s
=> naming to docker.io/sahanaa44/flask-k8s-demo:1.0                             0.0s

PS C:\Users\sahanaa\flask-k8s-demo> docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
sahanaa44/flask-k8s-demo  1.0         7e3c8b9f9a1a   10 seconds ago  152MB

PS C:\Users\sahanaa\flask-k8s-demo> docker push sahanaa44/flask-k8s-demo:1.0
The push refers to repository [docker.io/sahanaa44/flask-k8s-demo]
a1b2c3d4e5f6: Pushed
b1c2d3e4f5a6: Pushed
c1d2e3f4a5b6: Pushed
1.0: digest: sha256:9a1b7c6d5e4f3210fedcba0876543210abcd f1234567890fedcba0876543210 size: 1154

PS C:\Users\sahanaa\flask-k8s-demo> kubectl apply -f deployment.yaml
deployment.apps/flask-api created
PS C:\Users\sahanaa\flask-k8s-demo> kubectl apply -f service.yaml
service/flask-service created
PS C:\Users\sahanaa\flask-k8s-demo> kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/flask-api-6d7c8f4c7d-3k8xq     1/1     Running   0          11s
pod/flask-api-6d7c8f4c7d-lm9zj     1/1     Running   0          11s

NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/flask-service               NodePort       10.105.87.234   <none>            80:30080/TCP     11s

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/flask-api           2/2     2             2           11s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/flask-api-6d7c8f4c7d  2         2         2       11s

PS C:\Users\sahanaa\flask-k8s-demo> minikube ip
192.168.49.2
PS C:\Users\sahanaa\flask-k8s-demo>
```